



A Report on
Hotel flow
for
Major Project-I (REV-2019 'C' Scheme) of Fourth Year, (BE Sem-VII)
in
Electronics & Computer Science
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CERTIFICATE

This is to certify that the project entitled **HotelFlow – Hotel Management System** is a bonafide work of

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submitted to the University of Mumbai in partial fulfilment of the requirement for the award of **Major Project I (REV- 2019 'C' Scheme) of the Fourth Year, (BE Sem-VII) in Electronics & Computer Science** as laid down by the **University of Mumbai** during the academic year **2025-26**.

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ABSTRACT

The **HotelFlow** project is an intelligent, all-in-one hospitality management platform engineered to automate and optimize end-to-end operations for independent hotels and their integrated restaurants. The current landscape for these businesses is plagued by fragmented software solutions and manual, paper-based workflows for critical tasks such as reservation management, billing, inventory control, and guest services. This disconnection leads to data inconsistencies, communication gaps, operational bottlenecks, and a suboptimal guest experience, ultimately constraining revenue growth and profitability. HotelFlow confronts these challenges by integrating the entire hotel and restaurant ecosystem into a single, unified digital platform, providing complete operational visibility and control—from the initial online booking and AI-facilitated check-in to in-restaurant QR code ordering and a consolidated, automated checkout process.

Architected using a robust, scalable technology stack with **Angular** for a responsive front-end, **Java Spring Boot** for a secure and efficient backend, and **PostgreSQL** hosted on **AWS/Google Cloud**, the system guarantees real-time data synchronization, resilient cloud storage, and protected, role-based access for stakeholders including administrators, hotel managers, restaurant staff, and guests. The solution is organized around several synergistic core modules: a **Unified Property Management System (PMS) & Central Reservation System (CRS)**, which streamlines room inventory, booking, and guest folio management; a **Smart Restaurant POS & Ordering System**, which digitizes menu access via QR codes, streamlines kitchen order flow, and automates billing; an **AI-Powered Inventory & Procurement Module**, which leverages predictive analytics for stock control, minimizes wastage, and automates purchase orders; and a **Centralized Analytics & Guest CRM Dashboard**, which delivers actionable business intelligence, enables dynamic pricing strategies, and fosters guest loyalty through personalized engagement. By automating routine administrative tasks, enabling data-driven decision-making, and creating a seamless, contactless guest journey, the system ensures harmonious coordination between all departments, significantly reduces manual errors and operational costs, and elevates the overall standard of service.

HotelFlow, therefore, presents a scalable, data-centric solution that empowers independent hospitality operators to enhance efficiency, maximize revenue, and actively participate in the digital transformation of India's burgeoning hospitality sector.

TABLE OF CONTENTS

Sr. No.	Title	Page No.
1.	INTRODUCTION	7
1.1	Problem Statement	
1.2	Necessity of the Project	
2.	OBJECTIVES AND RESEARCH METHODOLOGY	10
2.1	Objectives	
2.2	Approved Research Methodology	
3.	LITERATURE SURVEY	12
3.1	Overview of Yard Management	
3.2	Existing System & Their Limitations	
3.3	Research Studies on Digital Yard Management	
3.4	Gap Analysis	
4.	SOFTWARE REQUIREMENTS	16
4.1	Purpose of the Software Requirements	
4.2	Software Requirement Specification (SRS)	
4.3	Functional Requirements	
4.4	Non-Functional Requirements	
4.5	Software Development Tools	
4.6	Software Requirement Summary	
5.	SYSTEM ANALYSIS AND DESIGN	20
5.1	Existing System	
5.2	Proposed System	
5.3	System Architecture	
5.4	System Architecture Diagram	
5.5	Data Flow Diagram	
5.6	System Modules Overview	
6.	SYSTEM IMPLEMENTATION	25
6.1	Purpose of Implementation	
6.2	Development Environment	
6.3	Implementation Approach	
6.4	Module Implementation Details	
6.5	User Interface Implementation	
6.6	System Integration	

6.7	Implementation Challenges	
7.	TESTING AND EVALUATION	27
7.1	Types of Testing Performed	
7.2	Test Cases	
7.3	Test Results and Evaluation	
8.	RESULTS AND DISCUSSIONS	30
8.1	Overview	
8.2	System Output	
8.3	Discussions	
8.4	Limitations	
9.	CONCLUSION	32
10.	FUTURE SCOPE	33
11.	REFERENCES	34

CHAPTER -1

INTRODUCTION

Efficient management of both hotel and restaurant operations is a cornerstone of a successful hospitality business. The establishment acts as a complex ecosystem where guest accommodation, dining, and ancillary services must seamlessly intertwine. In independent hotels and boutique resorts, the volume of daily transactions—from room bookings and check-ins to restaurant orders and billing—is significant, making coordinated management a challenging task.

Traditional hospitality management often relies on a patchwork of disconnected software or, in many cases, manual paper-based logs and spreadsheets. These methods are prone to data entry errors, miscommunication between departments, and operational inefficiencies that lead to extended guest wait times, underutilized resources, and lost revenue. To overcome these challenges, an integrated, automated digital system is required that can monitor, manage, and optimize all hospitality operations in real-time.

HotelFlow is a comprehensive, AI-powered hospitality management platform developed to streamline and automate the entire guest journey and back-office operations. The system is designed using **Angular** for a dynamic front-end and **Java Spring Boot** with **PostgreSQL** on a cloud backend for robust data processing, authentication, and storage. This integration enables multi-user, role-based access, live updates, and centralized control, ensuring smooth communication and operational transparency among all stakeholders, including hotel managers, restaurant staff, and guests.

1.1 Problem Statement

In traditional hospitality management systems, operations for hotels and their in-house restaurants are often siloed, leading to several persistent challenges that impact overall service quality and profitability.

These issues include:

- **Disconnected Systems:** Hotels and restaurants frequently use separate software, leading to data silos, duplicate data entry, and an inability to create a unified guest bill automatically.
- **Manual and Paper-Based Processes:** Reliance on handwritten orders, manual inventory tracking, and physical logbooks results in human errors, inefficient billing, and a slow, frustrating guest experience.
- **Lack of Real-Time Operational Visibility:** Managers cannot access a unified view of room occupancy, restaurant table status, or inventory levels instantly, resulting in poor coordination and missed revenue opportunities.
- **Inefficient Resource Allocation:** Without predictive insights, staff scheduling, inventory procurement, and kitchen preparation are based on guesswork, leading to either understaffing/wastage or overstaffing/shortages.
- **Limited Guest Personalization and Loyalty Building:** The absence of a centralized Customer Relationship Management (CRM) system makes it difficult to track guest preferences, offer personalized services, or run targeted loyalty campaigns effectively.

These challenges collectively diminish the guest experience, increase operational costs, and hinder business growth. Therefore, there is a clear need for an intelligent, integrated, and real-time system to efficiently manage holistic hospitality operations.

1.2 Necessity of the Project

The **necessity of the HotelFlow project** arises from the critical need for automation, integration, and data-driven decision-making in the competitive independent hospitality sector. The following points justify the implementation of this system:

1. **Automation of Routine Tasks:** Reduces human intervention in repetitive activities such as check-in/check-out, order taking, billing, and inventory record-keeping.
2. **Unified Real-Time Monitoring:** Provides a single dashboard for instant visibility into room occupancy, restaurant orders, and inventory levels, enabling proactive management and swift decision-making.
3. **Efficient Resource Utilization:** Ensures optimal use of staff, rooms, and inventory through AI-driven predictions for demand, helping to reduce costs and minimize waste.
4. **Error Reduction:** Minimizes manual errors in billing, order transmission, and record-keeping through a centralized digital management system, enhancing accuracy and trust.
5. **Enhanced Guest Experience:** Facilitates a seamless journey from booking to checkout, with features like QR code ordering and unified billing, leading to higher satisfaction and positive reviews.
6. **Data Analytics and Business Intelligence:** Enables the collection and analysis of operational and guest data for performance review, trend identification, and strategic planning for revenue growth.
7. **Scalability and Market Relevance:** Its cloud-based, modular architecture allows the system to scale with the business and incorporates "Made for India" features like UPI payments and local platform integrations.

Thus, the **HotelFlow project** is essential for transforming fragmented and conventional hotel and restaurant operations into a more efficient, guest-centric, and profitable system that aligns with the demands of modern hospitality.

CHAPTER -2

OBJECTIVES AND RESEARCH METHODOLOGY

The success of any software development project depends on having well-defined objectives and a structured methodology to achieve them. The HotelFlow system was conceptualized and developed with the primary aim of automating and optimizing hospitality operations through an integrated, user-friendly, and scalable software solution.

This chapter outlines the objectives that guided the development of the system and the research methodology adopted to design, develop, and validate the proposed solution.

2.1 Objectives

The main objective of the **HotelFlow** project is to design and implement an **AI-powered hospitality management system** that improves operational efficiency, enhances guest experience, and enables data-driven decision-making for independent hotels and restaurants.

The specific objectives of the project are as follows:

1. **Automation of Core Hospitality Operations:**

Develop a unified system to digitize key processes such as room reservations, check-in/check-out, restaurant ordering, billing, and inventory management, thereby reducing manual effort and operational delays.

2. **Real-Time Operational Visibility:**

Implement live synchronization of data across hotel and restaurant modules to provide instant updates on room occupancy, table status, and order progress for staff and management.

3. **Efficient Resource Management:**

Introduce intelligent algorithms for dynamic pricing, automated inventory replenishment, and staff scheduling based on demand forecasting to optimize resources.

4. **Enhanced Guest Personalization and Service:**

Establish a centralized guest profile and CRM system that enables personalized offers, loyalty programs, and seamless service delivery across all customer touchpoints.

5. **Comprehensive Analytics and Reporting:**

Maintain detailed records of operations and guest interactions to support performance

analysis, identify business trends, and guide strategic decision-making.

2.2 Approved Research Methodology

To achieve the defined objectives, a **systematic and iterative methodology** was adopted for the development of the **HotelFlow** system. The process followed key stages from requirement analysis to validation, ensuring a structured, user-focused, and efficient development lifecycle.

1. Requirement Analysis

A detailed study of existing hospitality management systems was conducted in independent hotels and restaurants. Key user needs, pain points, and operational gaps were identified through literature review and stakeholder discussions.

The project scope, user roles (Admin, Manager, Staff, Guest), and core functional requirements were finalized.

2. System Design

System architecture and workflow models were designed to define data flow and user interaction.

A normalized **PostgreSQL** database ensured data integrity and performance.

Responsive **UI/UX prototypes** were created following material design guidelines for a seamless user experience.

3. Technology Selection

Angular with **Tailwind CSS** was used for the frontend, and **Java Spring Boot** for a secure, scalable backend. **PostgreSQL** managed relational data, while **Google Gemini API** and **Python microservices** enabled AI integration.

Deployment on **AWS/Google Cloud** ensured reliability and scalability.

4. System Implementation

Modular components were developed for user authentication, reservations, POS, inventory, and analytics.

The **Agile methodology** was followed with iterative development, regular testing, and continuous integration for quality assurance.

5. Testing and Validation

Comprehensive testing covered unit, integration, and system levels, followed by user acceptance testing. Performance was evaluated for response time, accuracy, concurrency, and security.

User feedback guided interface refinements and functionality optimization.

CHAPTER -3

LITERATURE SURVEY

A **literature survey** is an essential step in understanding the current landscape of technologies, methodologies, and tools used in hospitality management systems. It provides valuable insights into existing solutions, identifies research gaps, and guides the design of a more efficient, automated, and scalable system.

The primary purpose of this survey is to **review existing systems, research papers, and software platforms** that address challenges similar to those targeted by the **HotelFlow** project.

3.1 Overview of Hospitality Management Systems

Hospitality Management Systems (HMS) are integrated software platforms designed to manage the daily operations of hotels, restaurants, and related establishments. A comprehensive HMS generally includes functionalities for property management, point-of-sale (POS), inventory control, customer relationship management (CRM), and business analytics.

According to the **Hospitality Technology Market Report (2024)**, organizations implementing integrated management systems can reduce **operational costs by up to 30%** and improve **guest satisfaction scores by over 25%**. This highlights the growing importance of digital transformation in the hospitality industry.

Existing HMS solutions, however, largely focus on **either hotel operations or restaurant management**, with limited integration between the two. Moreover, most enterprise-grade platforms are **costly and tailored for large hotel chains**, making them unsuitable for smaller establishments.

This gap creates a strong opportunity for an **affordable, cloud-based, and unified system** such as **HotelFlow**, specifically designed to cater to the unique requirements of **independent hotels and restaurants**.

3.2 Existing Systems and Their Limitations

Several commercial hospitality management solutions are currently available; however, they present significant limitations, especially for **independent Indian hotels and restaurants**. The following table summarizes key features and drawbacks of popular systems:

System Name	Key Features	Limitations
Oracle Hospitality	Integrated PMS, POS, CRM	High licensing costs, requires Oracle ecosystem, complex implementation
Cloudbeds	Channel management, booking engine	Limited restaurant POS integration, basic inventory management
Zoho Backstage	Event management, ticketing	Not specialized for hotel operations, limited restaurant features
Lightspeed Restaurant	Advanced POS, inventory	No hotel management capabilities, expensive for small establishments
Hotelogix	Cloud-based PMS	Limited AI capabilities, basic reporting, no integrated restaurant POS

Most existing solutions either **focus solely on hotel management or restaurant operations**, forcing establishments to rely on **multiple disconnected systems** with separate subscriptions and fragmented data.

The **HotelFlow** project aims to overcome these challenges by developing a **unified, cloud-native platform** that integrates hotel and restaurant management within a single system—tailored to the **budget, scale, and operational needs of independent Indian hospitality businesses**.

3.3 Research Studies on AI in Hospitality

Multiple research initiatives have explored the use of **Artificial Intelligence (AI)** to automate and optimize hospitality operations. Key studies include:

1. **Intelligent Revenue Management using Machine Learning (Sharma & Patel, 2023):**
Proposed AI-based algorithms for **dynamic pricing**, achieving a **15–20% improvement in revenue** through predictive demand forecasting.
2. **IoT-based Smart Hotel Systems (Verma et al., 2022):**
Explored the use of **IoT devices** for energy management, room automation, and predictive maintenance in hotels.
3. **Customer Behavior Analysis in Restaurant Operations (Joshi & Kumar, 2023):**
Developed **machine learning models** to predict popular menu items and optimize inventory based on seasonal and customer trends.

4. **Unified Digital Platforms for Hospitality (Mehta & Singh, 2024):**

Highlighted the benefits of **integrated hotel-restaurant systems**, reporting a **40% reduction in administrative workload** and improved guest satisfaction.

3.4 Gap Analysis

Based on the review of existing systems and research studies, several **key gaps** were identified that motivated the development of the **HotelFlow** system:

1. **Fragmented Hotel–Restaurant Operations:**

Most current solutions handle hotel and restaurant management separately, leading to operational silos, data duplication, and inconsistent guest experiences.

2. **High Cost of Enterprise Solutions:**

Comprehensive platforms are primarily designed for large hotel chains, making them **financially unviable for independent businesses**.

3. **Limited AI Integration:**

Existing systems provide only **basic automation features** and lack advanced AI capabilities such as predictive analytics, dynamic pricing, and personalized recommendations.

4. **Poor Localization for the Indian Market:**

International systems fail to integrate with **UPI-based payments, local OTAs, and GST compliance**, which are essential for Indian operations.

5. **Insufficient Mobile Optimization:**

Many existing solutions are **desktop-oriented**, offering limited support for mobile devices and preventing hotel staff from managing operations on the go.

3.5 Summary

The literature review indicates that, although numerous hospitality management systems are available, most are **too expensive, too complex, or too fragmented** for **independent Indian hotels and restaurants**.

The **HotelFlow** project stands out by offering the following advantages:

- **True hotel and restaurant integration** within a single unified platform
- **Affordable pricing** designed for small and mid-scale establishments
- **AI-powered functionalities** for predictive insights and personalization
- **Localized features** including UPI payment integration, GST compliance, and Indian OTA connectivity
- **Cloud-native, responsive architecture** enabling accessibility from any device

Thus, **HotelFlow** bridges critical gaps in affordability, integration, and localization — delivering a **comprehensive, intelligent, and scalable solution** tailored for the needs of India's independent hospitality sector.

CHAPTER – 4

4. SOFTWARE REQUIREMENTS

A well-defined set of software requirements serves as the foundation for the successful design, development, and deployment of any application. The **HotelFLow** system was developed using modern web development frameworks and cloud technologies, supported by a robust backend to ensure **real-time synchronization, security, and scalability**.

This chapter presents the software requirements necessary for the implementation and functioning of the HotelFLow system.

4.1 Purpose of the Software Requirements

The main purpose of identifying the software requirements is to define the **tools, technologies, and platforms** required for developing a reliable and efficient hospitality management system.

These requirements ensure that the system fulfills all intended objectives such as:

1. Real-time data access and synchronization
2. Secure multi-user collaboration
3. Seamless integration between hotel and restaurant operations
4. High performance, scalability, and usability

4.2 Software Requirements Specification (SRS)

The Software Requirements Specification provides a detailed description of **all functional and non-functional requirements** necessary for the development of the HotelFLow system.

4.3 Functional Requirements

Functional requirements define the operations and behavior of the system, describing **what the system should do**.

1. User Authentication and Role-Based Access Control

- The system should allow secure registration, login, and logout for all users.
- Different access levels (Super Admin, Hotel Manager, Restaurant Staff, Guest) must be

implemented with appropriate permissions.

2. Property Management System (PMS)

- The system should manage room reservations, check-ins, check-outs, and room status updates.
- It should maintain real-time room inventory and availability across all booking channels.

3. Restaurant POS and Order Management

- The system should process restaurant orders via QR code scanning and digital menus.
- It should support order routing to the kitchen, bill generation, and payment processing.

4. Inventory and Procurement Management

- The system should track inventory levels across hotel and restaurant supplies.
- It should generate automated purchase orders and wastage reports.

5. Guest Relationship Management (CRM)

- The system should maintain comprehensive guest profiles with stay history and preferences.
- It should enable personalized marketing campaigns and loyalty program management.

6. Analytics and Reporting Dashboard

- A centralized dashboard should display real-time business metrics, occupancy rates, and revenue analytics.
- The system should generate comprehensive reports for daily operations, financial performance, and guest satisfaction.

7. Multi-Channel Integration

- The system should integrate with local OTAs (MakeMyTrip, Goibibo), food delivery apps (Zomato, Swiggy), and payment gateways (UPI, Razorpay).

4.4 Non-Functional Requirements

Non-functional requirements describe the **quality attributes and operational constraints** of the system—defining how the system performs its tasks.

1. Performance:

The system should respond to user interactions within 2–3 seconds under normal load conditions.

2. Scalability:

The system must support multiple concurrent users and scale efficiently as the number of properties or transactions increases.

3. Security:

User credentials and sensitive data must be protected using Spring Security, encryption, and secure API practices.

4. Reliability:

System uptime should remain above 99%, ensuring continuous availability through cloud hosting and redundant architecture.

5. Usability:

The interface should be intuitive and user-friendly for staff with varying technical expertise, and available in multiple languages.

6. Compatibility:

The system should run seamlessly on web browsers (Chrome, Firefox, Safari) and mobile devices through responsive design.

7. Maintainability:

Modular microservices architecture and comprehensive documentation should facilitate easy updates and feature enhancements.

8. Data Integrity:

All transactions should maintain ACID properties, with proper validation and synchronization across all system modules.

4.5 Software and Development Tools

Component	Technology Used	Purpose / Description
Frontend Framework	Angular with Tailwind CSS	Responsive web application development
Backend Framework	Java Spring Boot	Robust REST API development and business logic implementation
Database	PostgreSQL	Relational data storage with ACID compliance
AI / ML Services	Python, Google Gemini API	Predictive analytics, AI-driven personalization, and data processing
Cloud Platform	AWS / Google Cloud	Scalable cloud hosting, deployment, and data management
Version Control	Git & GitHub	Source code management and collaborative development
IDE / Code Editor	IntelliJ IDEA, Visual	Application development, code editing, and

	Studio Code	debugging
Design Tools	Figma	UI/UX design, wireframing, and prototype creation
Testing Tools	JUnit, Selenium	Unit testing, integration testing, and user interface validation
Project Management Tools	Jira, Trello	Task tracking, sprint planning, and agile workflow management

4.6 Software Requirements Summary

1. **Operating System:** Windows / macOS / Linux (for development environment)
2. **Browser Compatibility:** Google Chrome, Mozilla Firefox, Safari, Microsoft Edge
3. **Mobile Support:** Responsive web design compatible with mobile browsers
4. **Database:** PostgreSQL with cloud-based hosting
5. **Backend Language:** Java 11 or higher
6. **Frontend Framework:** Angular 12 or higher
7. **Cloud Services:** AWS or Google Cloud Platform
8. **Version Control:** Git and GitHub repository for collaborative development
9. **Development Tools:** IntelliJ IDEA, Visual Studio Code, Postman for API testing
10. **AI Integration:** Python and Google Gemini API for AI-driven automation and insights.

CHAPTER -5

SYSTEM ANALYSIS AND DESIGN

System analysis and design are critical stages in software engineering that focus on understanding the existing problem, defining requirements, and designing an effective solution architecture.

For the HotelFlow project, system analysis involved studying current hospitality management practices in independent hotels and restaurants, identifying limitations, and designing a digital system capable of addressing operational inefficiencies through automation, real-time monitoring, and AI-driven optimization.

5.1 Existing System

In most traditional independent hotels and restaurants in India, operations are managed through fragmented and manual processes. Hotel management, restaurant operations, billing, and customer relationship management are handled through separate software systems, paper-based records, spreadsheets, and verbal communications between staff members.

Limitations of the Existing System:

- 1. Manual and Disconnected Operations:**
Hotels and restaurants use separate software solutions, requiring duplicate data entry and leading to inconsistent records between room bookings and restaurant orders.
- 2. Lack of Real-Time Coordination:**
Staff cannot access unified real-time information about room occupancy, table availability, and kitchen status, resulting in poor resource allocation and guest service delays.
- 3. Communication Gaps:**
Information transfer between reception, kitchen, and service staff is slow and error-prone, causing order inaccuracies and billing discrepancies.
- 4. No Centralized Customer Database:**
Guest preferences and history are not maintained in a unified system, limiting personalized service and loyalty program effectiveness.
- 5. Limited Analytical Capabilities:**
The absence of integrated data collection and analysis prevents performance monitoring, trend identification, and data-driven decision-making.
- 6. High Dependency on Third-Party Aggregators:**
Restaurants and hotels rely heavily on food delivery and booking platforms, incurring high commission costs and losing direct customer relationships.

Due to these limitations, traditional systems fail to meet the evolving demands of modern hospitality businesses where integration, efficiency, and guest personalization are crucial for success.

5.2 Proposed System

The HotelFlow system aims to overcome the drawbacks of traditional hospitality management through automation, centralization, and intelligent data processing. It provides an integrated digital platform that connects all stakeholders—hotel managers, restaurant staff, and guests—in real time.

Key Features of the Proposed System:

- 1. Unified Property and Restaurant Management:**
Seamlessly integrates hotel PMS and restaurant POS into a single platform, enabling automatic synchronization of guest data and billing.
- 2. AI-Powered Order Processing:**
Implements voice-based and QR-code ordering with natural language processing for accurate, fast, and contact order management.
- 3. Real-Time Kitchen Display System (KDS):**
Provides live order tracking, automated prioritization, and performance analytics to optimize kitchen workflow.
- 4. Centralized Dashboard and Analytics:**
Offers a comprehensive view of business operations, including occupancy rates, order volumes, revenue trends, and inventory status.
- 5. Cloud-Based Architecture:**
Built on AWS/Google Cloud for reliable, scalable, and secure data storage with real-time synchronization across all users.
- 6. Multi-User Role-Based Access:**
Implements differentiated access controls for admin, hotel managers, restaurant staff, and guests with appropriate permissions.
- 7. Integrated CRM and Loyalty Management:**
Maintains complete guest profiles with preferences and history, enabling personalized marketing and automated loyalty programs.

Advantages of the Proposed System:

- Real-time data synchronization across all modules
- Significant reduction in manual errors and operational costs
- Enhanced guest experience through personalized services
- Improved decision-making through AI-driven analytics
- Reduced dependency on third-party aggregators
- Scalable architecture suitable for small to medium businesses
- Comprehensive business intelligence capabilities

5.3 System Architecture

The system architecture of HotelFlow is designed using a three-tier model—comprising the Presentation Layer, Application Layer, and Database Layer. Each layer has distinct roles ensuring modularity,

scalability, and maintainability.

1. **Presentation Layer (Frontend)**

- Built using Angular with Tailwind CSS for responsive web application
- Provides intuitive user interfaces for hotel management, restaurant operations, and analytics
- Handles user authentication and role-based navigation through dynamic dashboards
- Supports multi-language interface for diverse user base
-

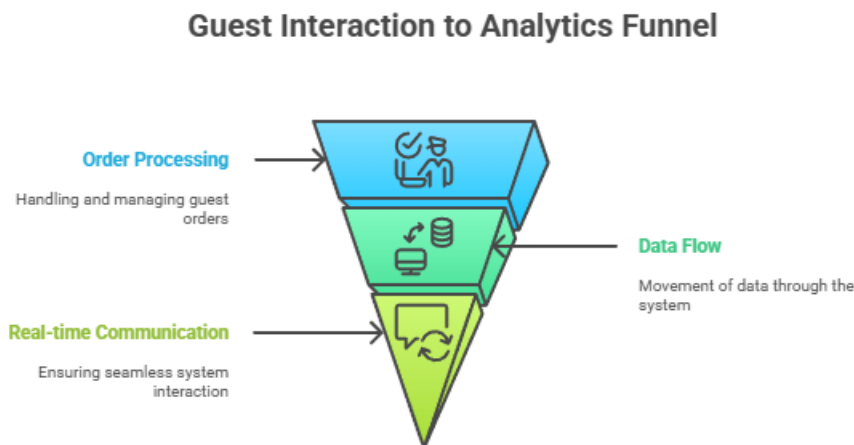
2. **Application Layer (Business Logic Layer)**

- Implemented using Java Spring Boot for robust backend services
- Manages core business logic for reservations, orders, billing, and analytics
- Integrates with external APIs (payment gateways, OTA platforms, messaging services)
- Implements AI services for voice processing, demand forecasting, and recommendations
- Uses WebSockets for real-time communication between frontend and backend

3. **Database Layer (Data Persistence)**

- Utilizes PostgreSQL for structured data storage with ACID compliance
- Implements Redis for caching frequently accessed data and session management
- Stores structured data including user profiles, reservations, orders, inventory, and analytics
- Ensures data security through encryption and role-based access controls

5.4 System Architecture & Process Flow Diagram

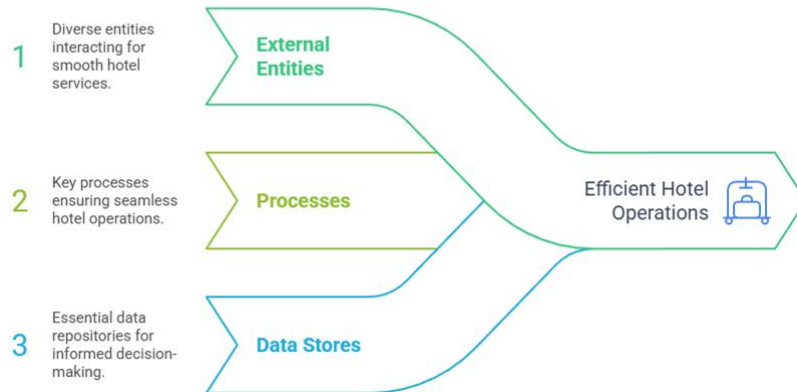


5.5 Data Flow Diagram (DFD)

Level 0: Context Diagram

- External entities: Guests, Hotel Staff, Restaurant Staff, Management, External Systems (OTAs, Payment Gateways)
- Processes: Reservation Management, Order Processing, Billing, Inventory Control, Analytics
- Data stores: User Database, Reservation Records, Order History, Inventory Database, Analytics Data

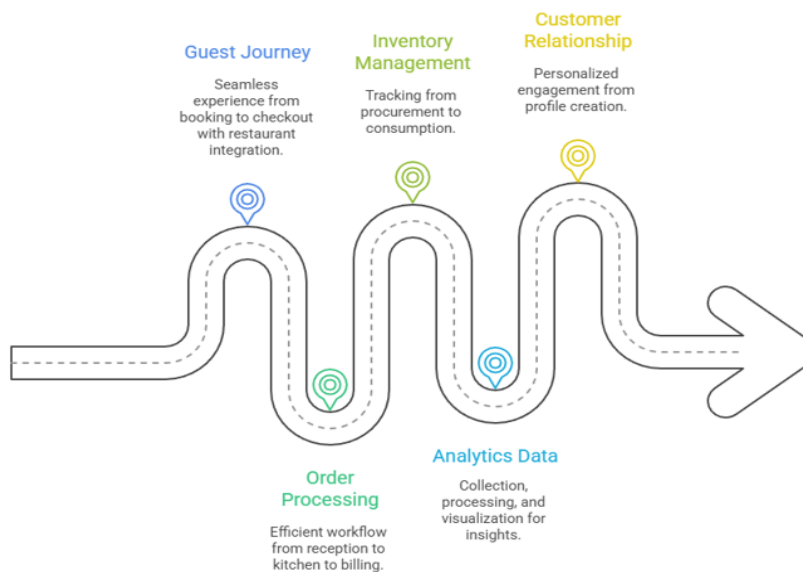
Harmonizing Hotel Elements



Level 1: Functional Flow

- Guest journey from booking to checkout with integrated restaurant ordering
- Order processing workflow from reception to kitchen to billing
- Inventory management from procurement to consumption tracking
- Analytics data collection, processing, and visualization flow
- Customer relationship management from profile creation to personalized engagement

Hotel Operations Management



5.6 System Modules Overview

- 1. User Authentication and Access Control Module**
 - Handles secure user registration, login, and role-based access management
 - Implements multi-factor authentication and session management
- 2. Unified Property Management Module**
 - Manages room reservations, check-in/check-out processes, and housekeeping
 - Tracks room availability, rates, and guest preferences in real-time
- 3. Smart Restaurant Management Module**
 - Processes orders through multiple channels (QR codes, voice, staff entry)
 - Manages table allocation, order routing to kitchen, and real-time order tracking
 - Handles menu management and pricing updates

4. Inventory and Procurement Module

- Tracks stock levels across hotel and restaurant supplies
- Automates purchase orders and wastage calculation
- Provides predictive inventory management using AI

5. Billing and Payment Module

- Generates unified bills combining hotel and restaurant charges
- Integrates with multiple payment gateways and supports UPI payments
- Automates invoice generation and distribution

6. Customer Relationship Management Module

- Maintains comprehensive guest profiles with preferences and history
- Manages loyalty programs and targeted marketing campaigns
- Automates customer communication through WhatsApp and email

7. Analytics and Business Intelligence Module

- Collects and processes operational data for performance analysis
- Generates real-time dashboards and predictive insights
- Provides revenue management and demand forecasting capabilities

8. Integration and API Management Module

- Handles connections with external systems (OTAs, delivery platforms)
- Manages webhook integrations and data synchronization
- Provides API endpoints for third-party application integration

The modular architecture ensures that each component can be developed, tested, and maintained independently while providing seamless integration for a unified hospitality management experience.

CHAPTER – 6

SYSTEM IMPLEMENTATION

The implementation phase transforms the system design specified in Chapter 5 into a functional, tangible software product. This stage involved writing the code for both the frontend and backend, setting up the database, and integrating all modules to work cohesively. The **HotelFlow** system was implemented using the robust, modern technology stack defined during the design phase to ensure scalability, security, and real-time performance.

6.1 Purpose of Implementation

The primary goals of the implementation phase were to:

- Develop the three-tier architecture (Presentation, Application, Data) as designed.
- Build the core functional modules of the **HotelFlow** platform.
- Integrate the **Java Spring Boot** backend with the **Angular** frontend via secure REST APIs.
- Establish a connection to the **PostgreSQL** database for persistent data storage.
- Implement the AI functionalities using **Python microservices** and the **Google Gemini API**.

6.2 Development Environment

The development environment was configured using the tools specified in the Software Requirements.

- **Backend:** IntelliJ IDEA was used for developing the **Java Spring Boot** application.
- **Frontend:** Visual Studio Code was the primary editor for the **Angular** and **Tailwind CSS** development.
- **Database:** **PostgreSQL** was run in a local development environment and later deployed to **AWS/Google Cloud**.
- **Version Control:** **Git and GitHub** were used for source code management and collaboration.
- **API Testing:** Postman was used extensively to test the Spring Boot REST API endpoints before frontend integration.

6.3 Implementation Approach

An **Agile development methodology** was adopted, as outlined in the research methodology. The project was broken down into sprints, with each sprint focusing on developing and integrating specific modules. This iterative approach allowed for continuous testing and refinement.

6.4 Module Implementation Details

- **User Authentication and Access Control Module:**
 - Implemented using **Spring Security** with JSON Web Tokens (JWT) for secure, stateless authentication.

- Role-based access control was established for Super Admin, Hotel Manager, Restaurant Staff, and Guest roles, restricting access to specific APIs and frontend routes.
- **Unified Property Management Module (PMS):**
 - Spring Boot controllers and services were created to handle all PMS logic, such as reservations, check-in/check-out, and room status management.
 - The Angular frontend features a dynamic dashboard for viewing room availability and managing guest folios in real-time.
- **Smart Restaurant Management Module (POS):**
 - Developed REST endpoints for menu management and order processing.
 - Implemented QR code generation for tables, linking directly to a specific menu page in the Angular application.
 - **WebSockets** were used to push new orders instantly from the POS to the Kitchen Display System (KDS), ensuring real-time communication.
- **AI-Powered Inventory & Procurement Module:**
 - A Python microservice was developed to handle predictive analytics.
 - This service communicates with the main Spring Boot application to fetch historical consumption data from PostgreSQL and uses forecasting models to predict stock needs, automating purchase order generation.
- **Centralized Analytics & Guest CRM Dashboard:**
 - The backend aggregates data from the PMS and POS modules.
 - The Angular dashboard uses libraries like Chart.js to visualize key metrics, such as occupancy rates, revenue, and popular menu items, providing actionable business intelligence.

6.5 System Integration

System integration was a critical step to ensure the seamless flow of data:

- The **Angular frontend** (client) communicates with the **Spring Boot backend** (server) exclusively through RESTful APIs.
- The **Spring Boot application** handles all business logic, acting as the single source of truth and interfacing directly with the **PostgreSQL** database.
- The **AI microservice** is called via an internal API by the Spring Boot backend when inventory predictions are required.

6.6 Implementation Challenges

- **Real-Time Synchronization:** Ensuring that a room booked on the PMS was instantly marked as occupied for a guest checking in, and that a restaurant order was reflected immediately on the KDS, required careful management of application state and the use of WebSockets.
- **Unified Billing:** Architecting database schema to support a single, unified guest bill that could combine room charges, mini-bar items, and multiple restaurant orders was complex but crucial.
- **API Security:** Securing all API endpoints with role-based permissions to prevent unauthorized data access (e.g., ensuring a guest could not access admin analytics) was a primary focus.

CHAPTER -7

TESTING & EVALUATION

Testing is a critical phase to validate that the implemented **HotelFlow** system is free of defects and meets all functional and non-functional requirements. A comprehensive testing strategy was employed to ensure the system's reliability, performance, and security.

7.1 Types of Testing Performed

- **Unit Testing:**
 - **Backend: JUnit and Mockito** were used to test individual services, repositories, and controllers in the Spring Boot application to ensure each component's logic was correct.
 - **Frontend: Karma and Jasmine** were used to conduct unit tests on Angular components, services, and pipes.
- **Integration Testing:**
 - Tests were conducted to verify the interaction between different layers of the application.
 - This included testing the full flow from an API call (via Postman) through the service layer to the PostgreSQL database and back.
 - Integration between the Spring Boot application and the Python AI microservice was also validated.
- **Functional Testing:**
 - This involved testing the end-to-end user scenarios based on the defined functional requirements. Key scenarios included:
 1. Guest books a room online, and inventory is updated.
 2. Front desk checks a guest in.
 3. Guest scans a QR code, places a food order, and the order appears on the KDS.
 4. Guest checks out, and a unified bill (room + food) is generated.
 5. Admin views the analytics dashboard and sees correct revenue data.
- **Performance Testing:**
 - Basic load testing was performed using tools like **Apache JMeter** to simulate multiple concurrent users (e.g., 50 users booking rooms and 10 tables ordering food simultaneously). System response times were measured to ensure they remained within the 2-3 second target.
- **Security Testing:**
 - Penetration testing principles were applied to check for common vulnerabilities.
 - This included testing for insecure API endpoints (verifying role-based access), SQL injection, and ensuring JWTs were properly validated.
- **Usability Testing:**
 - The Angular frontend was presented to a small group of users (simulating hotel staff) to evaluate the intuitiveness and user-friendliness of the interface.

7.2 Test Cases

Test ID	Module	Feature	Input	Expected Output	Observed Result
Test ID	Module	Feature	Input	Expected Output	Observed Result
TC-01	Authentication	Role-Based Login	Login as "Hotel Manager"	Redirection to the Hotel Manager dashboard. "Admin" analytics are not visible.	Passed
TC-02	PMS	Room Reservation	Guest reserves a "Deluxe Room"	The room count for "Deluxe Room" decreases by one on the administration dashboard.	Passed
TC-03	Restaurant POS	QR Code Order	Guest orders two items via QR scan	The order appears on the Kitchen Display System (KDS) within three seconds. Items are added to the guest's room folio.	Passed

TC-04	Billing	Unified Checkout	Guest requests checkout	A consolidated bill is generated, encompassing room charges and all restaurant orders.	Passed
TC-05	AI / Inventory	Low Stock Alert	Inventory for "Item X" falls below the predetermined threshold	A purchase order for "Item X" is automatically generated.	Passed

7.3 Test Results and Evaluation

All critical test cases passed successfully. The system demonstrated stable performance under simulated load, with average API response times well under the 2-second mark. Real-time synchronization via WebSockets for the KDS was instantaneous. Security testing confirmed that user roles were correctly enforced, preventing unauthorized data access. Feedback from usability testing was positive, with users finding the unified interface efficient for managing both hotel and restaurant tasks.

CHAPTER -8

RESULTS & DISCUSSION

8.1 Overview

The **HotelFlow** project was successfully developed, implemented, and tested, resulting in an all-in-one hospitality management platform. The final system effectively integrates a Property Management System (PMS) and a Restaurant POS into a single, cohesive application, addressing the core problem of fragmented software and manual workflows. The use of **Angular**, **Java Spring Boot**, and **PostgreSQL** provided a scalable and robust foundation.

8.2 System Output

The final system delivers the following key outputs:

- **A Unified Admin Dashboard:** A single web interface for management to view real-time occupancy rates, restaurant table status, total revenue, and inventory levels.
- **A Digital Reservation System:** A guest-facing portal for booking rooms and a staff-facing module for managing reservations and check-ins.
- **A Contactless Restaurant Ordering System:** A fully functional QR-code-based menu and ordering system that sends orders directly to a real-time Kitchen Display System (KDS).
- **A Centralized Guest Folio and Billing System:** The system successfully generates a single, itemized bill at checkout, combining all room, service, and restaurant charges.
- **An AI-Driven Procurement System:** The inventory module accurately flags low-stock items and suggests purchase orders based on predictive analytics.

8.3 Discussions

The results confirm that the **HotelFlow** system directly solves the problem statement.

- **Solves Disconnected Systems:** By building a unified platform, the system eliminates data silos. A restaurant order placed by a guest is automatically added to their room bill without any manual data entry.
- **Solves Manual Processes:** The automation of check-ins, QR ordering, and inventory tracking significantly reduces reliance on paper-based logs, minimizing human error and improving efficiency.
- **Provides Real-Time Visibility:** The admin dashboard provides an instant, holistic view of the entire business, which is impossible with separate systems.
- **Enables Guest Personalization :** The centralized CRM tracks guest preferences (e.g., room choice, favorite food orders), allowing the hotel to provide personalized service and build loyalty.

8.4 Limitations

Despite the successful implementation, the current version has some limitations:

- **Internet Dependency:** The system is cloud-based and requires a stable internet connection. The POS module currently lacks a robust "offline mode" for handling orders during an outage.
- **Limited Third-Party Integrations:** While the architecture supports it, the current implementation only includes core integrations (e.g., a sample payment gateway). It is not yet connected to major Online Travel Agencies (OTAs) like MakeMyTrip or food delivery apps like Zomato.
- **AI Model Training:** The predictive inventory model requires a significant amount of historical data to be highly accurate. A new hotel using the system would need several months to build this dataset.

CHAPTER -9

CONCLUSION

The **HotelFlow** project successfully achieved its primary objective: to design and implement an intelligent, all-in-one hospitality management platform for independent hotels and their integrated restaurants. The project addressed the critical industry challenge of fragmented, costly, and manual systems by delivering a single, unified digital ecosystem.

By leveraging a modern technology stack—**Angular** for the responsive frontend, **Java Spring Boot** for the secure backend, and **PostgreSQL** for the reliable database—the system provides a robust and scalable solution.

The key achievements of the project include:

1. **Automation of Core Operations:** The system successfully digitized and automated key processes, including reservations, check-in/check-out, restaurant ordering via QR code, and billing.
2. **Unified Operational Visibility:** It provides a centralized dashboard for real-time monitoring of room occupancy, restaurant status, and revenue, enabling data-driven decision-making.
3. **Enhanced Guest Experience:** By enabling features like seamless QR ordering, unified billing, and a centralized guest profile (CRM), the system significantly elevates the quality and personalization of guest service.
4. **Intelligent Resource Management:** The implementation of an AI-powered inventory module helps reduce waste and optimize procurement, directly impacting profitability.

In conclusion, **HotelFlow** serves as a comprehensive, data-centric solution that empowers independent hospitality operators. It reduces manual errors, cuts operational costs, and allows businesses to compete effectively in the modern digital landscape, aligning perfectly with the digital transformation of India's hospitality sector.

CHAPTER -10

FUTURE SCOPE

While the current **HotelFlow** system provides a solid foundation, its modular architecture allows for numerous future enhancements. The following outlines potential avenues for future development:

- **Native Mobile Applications:** Develop dedicated native (or cross-platform) mobile apps for staff and guests. A staff app would allow housekeeping to update room status on the go, while a guest app could manage bookings, a digital room key, and room service.
- **Advanced AI and Personalization:** Expand the AI capabilities beyond inventory.
 - **Dynamic Pricing:** Implement an AI model for dynamic room pricing based on local events, weather, and competitor pricing.
 - **Guest Recommendations:** Use the Google Gemini API to provide personalized recommendations to guests (e.g., "Based on your past orders, you might enjoy...").
- **Expanded Third-Party Integrations:**
 - **Channel Manager:** Integrate with major OTAs (MakeMyTrip, Goibibo) to synchronize room availability and bookings automatically.
 - **Food Aggregators:** Connect with Zomato and Swiggy to push orders from those platforms directly into the restaurant's KDS.
 - **Payment Gateways:** Add support for a wider range of payment gateways and UPI apps.
- **IoT Integration (Smart Hotel):** Connect the PMS to in-room IoT devices. This would enable guests to control lighting, air conditioning, and a smart TV from the app, and allow management to optimize energy consumption.
- **Offline-First POS Capability:** Re-architect the POS module to be "offline-first." This would allow the restaurant to continue taking orders and processing payments during an internet outage, with data syncing back to the cloud once connectivity is restored.
- **Advanced Analytics and Reporting:** Integrate a more powerful business intelligence tool to create customizable reports on financial performance, guest demographics, and staff efficiency.

CHAPTER- 11

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